
Probabilistic Snowfall Forecasting at Ski Resorts

Sebastian Teslic
Rutgers University
slt158@rutgers.edu

Finn Ryan
Rutgers University
fmr48@rutgers.edu

Javier Martinez Alvarez
Rutgers University
jm1941@rutgers.edu

Abstract

We developed a machine learning project on probabilistic snowfall forecasting at ski resorts. Our goal was to predict both snowfall amount and snowfall occurrence probability using historical SNOTEL weather station observations and engineered atmospheric features. We compared baseline regression models, including Linear Regression and Random Forest, with a two-stage XGBoost forecasting model consisting of a snow/no-snow classifier and a conditional snowfall regressor. The model utilized temperature, precipitation, snow depth, lag features, rolling-window statistics, and seasonal indicators to learn temporal snowfall patterns. We evaluated predictive performance using RMSE, MAE, and classification metrics including precision, recall, and F1-score. Our results showed that the two-stage XGBoost approach achieved competitive snowfall forecasting performance relative to baseline models while effectively identifying snowfall events through probabilistic occurrence prediction. These findings suggest that uncertainty-aware and conditional machine learning approaches can improve short-term snowfall prediction for ski resort operations and environmental forecasting applications.

GitHub Repository: <https://github.com/steslic/cs439-project>

Project Video: <https://drive.google.com/file/d/1PBcpHJKn3RttQpPj7JR9SNUez5lUVsYb/view?usp=sharing>

1 Introduction and Related Work

Accurate snowfall prediction is important for ski resort operations and tourism. Resorts rely on snowfall forecasts to make decisions such as trail openings, staffing, visitor planning, and safety management. Because snowfall accumulation can vary significantly across storms and seasonal conditions, uncertainty-aware forecasting methods may provide more useful information than deterministic predictions alone. Commercial forecasting systems typically report a single expected snowfall value, while probabilistic forecasting approaches can additionally estimate forecast uncertainty and the likelihood of snowfall events.

Traditional snowfall forecasting has historically relied on weather simulation models and statistical forecasting methods. Predicting snowfall is challenging because snowfall accumulation depends on nonlinear interactions between temperature, precipitation, moisture, elevation, and seasonal weather conditions. Small temperature variations near the freezing point can affect snow-to-liquid ratios and overall snow accumulation, making accurate forecasting difficult.

Machine learning approaches have been applied to weather forecasting problems due to their ability to model nonlinear relationships and complex feature interactions. Methods such as XGBoost and Random Forest have been used in weather and precipitation forecasting because of their strong predictive performance. Time-series forecasting methods have also been used to model temporal weather dependencies and storm persistence using historical weather observations collected over consecutive days.

Prior forecasting models often focus primarily on deterministic point predictions rather than uncertainty-aware forecasting. Methods that estimate uncertainty are increasingly important in environmental forecasting applications where events and measurement variability can significantly affect forecast reliability. Unlike prior work focused only on deterministic snowfall prediction, our project combines probabilistic snowfall occurrence prediction with conditional snowfall amount regression using engineered weather and seasonal features. We also compare multiple baseline machine learning models to evaluate the effectiveness of a two-stage forecasting architecture.

2 Methodology

2.1 Data Collection and Preprocessing

We used historical SNOTEL weather station observations from the Grand Targhee, Wyoming region as the primary data source for snowfall forecasting. The dataset included daily measurements of snow water equivalent (SWE), precipitation accumulation, snow depth, and temperature variables. The raw data was parsed, cleaned, and chronologically sorted by date before preprocessing and feature engineering.

Missing values in snowfall, precipitation, snow depth, and temperature variables were handled using forward-filling techniques. Remaining incomplete rows were removed from the dataset. Temperature variables were converted from Fahrenheit to Celsius to support feature engineering and snowfall estimation. Because SNOTEL precipitation measurements are cumulative over the year, daily precipitation values were calculated using first differences, with negative resets clipped to zero.

Daily snowfall estimates were derived from positive changes in snow water equivalent. To estimate snowfall depth, we applied a temperature-based snow ratio approximation in which colder temperatures produced higher snowfall-to-SWE ratios. These snowfall estimates were smoothed using a rolling average to reduce short-term noise.

After preprocessing, additional weather-related features were generated for use in the forecasting models. These included lag variables for snowfall, precipitation, snow depth, and temperature using 1-day, 2-day, 3-day, 7-day, and 14-day windows. Additional features included temperature averages, temperature ranges, freezing indicators, seasonal cyclical encodings, and rolling-window statistics.

2.2 Feature Engineering

We engineered additional temporal and seasonal features including lag variables, rolling-window statistics, precipitation-temperature interaction features, freezing indicators, and cyclical seasonal encodings. Lag variables were generated for snowfall, precipitation, snow depth, and temperature in order to capture recent weather trends and storm persistence patterns over multiple days.

Rolling-window statistics including rolling means, rolling sums, and rolling maxima were computed to model snowfall accumulation and precipitation behavior across multiple days. Seasonal features were represented using month and day-of-year cyclical trigonometric encodings to preserve continuous seasonal relationships and recurring yearly weather behavior.

Temperature-based features including average temperature and temperature range were also included. Interaction features including precipitation-temperature interaction terms, along with freezing indicators, were used to help distinguish snowfall-producing conditions from non-snowfall-producing conditions.

2.3 Forecasting Models

We utilized several machine learning models for snowfall forecasting, including Linear Regression, Random Forest, and a two-stage XGBoost forecasting model. The baseline models allowed us to compare standard regression approaches against a conditional probabilistic forecasting framework.

Our final forecasting system consisted of two stages. First, an XGBoost classifier predicted the probability of snowfall occurrence for the next day. Second, an XGBoost regressor predicted snowfall amounts conditional on snowfall occurring. This approach allowed the model to separately learn snowfall occurrence and snowfall intensity patterns.

2.4 Training Procedure

Because snowfall forecasting is a time-series problem, the dataset was split chronologically in order to avoid temporal data leakage. The first 80% of observations were used for training and the remaining 20% were used for testing. The training set contained 5,702 observations while the test set contained 1,426 observations.

Baseline Linear Regression and Random Forest models were trained to provide comparative benchmarks against the two-stage XGBoost model. The two-stage XGBoost forecasting system first trained a binary classifier to predict whether snowfall would occur on the following day. Classification probabilities were converted into snowfall occurrence predictions using a tuned probability threshold. The classification threshold was tuned experimentally to balance precision and recall while minimizing false positive snowfall predictions.

The regression model was only used on days where the classifier predicted snowfall to estimate the snowfall amount. To reduce the influence of extreme snow variability, snowfall regression targets were log-transformed during training. Additionally, the classifier used class weighting to account for the unequal distribution of snowfall and non-snowfall observations.

3 Evaluation and Results

3.1 Evaluation Metrics

Model performance was evaluated using both regression and classification metrics. Regression performance was measured using root mean squared error (RMSE) and mean absolute error (MAE). Snowfall occurrence classification performance was evaluated using precision, recall, F1-score, and confusion matrices.

3.2 Regression Model Comparison

Table 1: Regression model comparison for snowfall forecasting performance.

Model	RMSE (inches)	MAE (inches)
Linear Regression	2.27	1.37
Random Forest	2.26	1.25
Two-Stage XGBoost	2.45	1.17

The regression model comparison results showed that both the Random Forest and two-stage XGBoost models achieved lower prediction error than the baseline Linear Regression model. The two-stage XGBoost model achieved the lowest MAE value of 1.17, indicating strong average snowfall prediction performance.

The Random Forest model achieved the lowest RMSE value of 2.26 compared to the XGBoost model’s RMSE of 2.45. This suggests that while the XGBoost model handled typical snowfall events effectively, larger snowfall events and extreme snowfall events remained more difficult to predict accurately.

3.3 XGBoost Classifier Metrics

Table 2: Two-stage XGBoost classifier performance metrics.

Metric	Value
Accuracy	85.48%
Precision	73.93%
Recall	60.70%
F1-score	66.7%

The two-stage XGBoost classifier achieved strong snowfall prediction performance, with an accuracy of 0.8548, precision of 0.7393, recall of 0.6070, and F1-score of 0.6667. Threshold tuning improved the balance between precision and recall by reducing false positive snowfall predictions while still correctly identifying a majority of snowfall events.

3.4 Visualization and Analysis

Exploratory data analysis and visualization techniques were also used to better understand model performance. This included snowfall distribution histograms, time-series prediction plots, confusion matrix visualizations, and actual vs predicted scatter plots. Additionally, feature importance analysis was performed to identify which engineered weather and seasonal variables contributed most to forecasting performance. Feature importance analysis showed that precipitation-temperature interaction variables contributed most strongly to snowfall occurrence prediction, supporting the importance of nonlinear environmental relationships in snowfall forecasting.

The full and clipped time-series snowfall prediction plot demonstrated that the two-stage XGBoost model successfully captured many seasonal snowfall patterns and especially moderate snowfall events across multiple winter seasons. The model correctly predicted many transitions between snowfall and non-snowfall periods and generally followed the timing of major snowfall periods. However, extreme snowfall events remained difficult to accurately predict; the model occasionally underestimated large snowfall spikes. The model produced few false snow predictions, indicating effective classification performance.

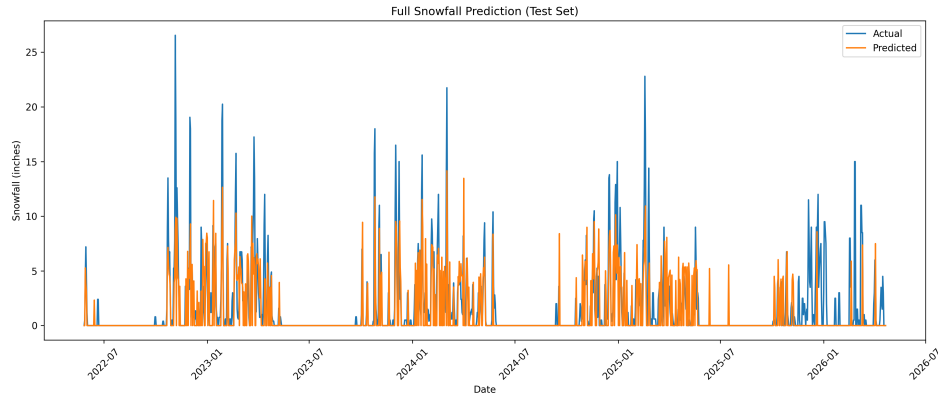


Figure 1: Time-series snowfall prediction results from the two-stage XGBoost model.

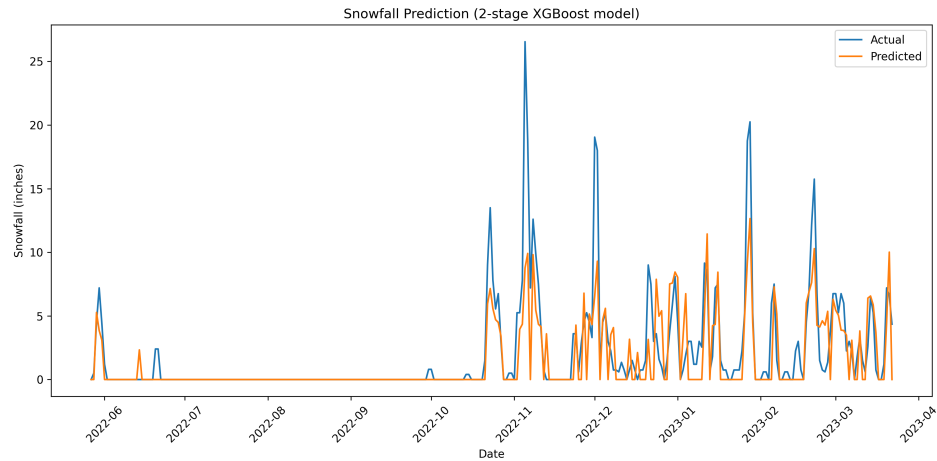


Figure 2: Time-series snowfall prediction results from the two-stage XGBoost model.

The snowfall distribution histogram revealed the heavily right-skewed, zero-inflated nature of daily snowfall at Grand Targhee. The majority of days recorded no meaningful snowfall and a long tail of infrequent large events. This distribution motivated the two-stage forecasting framework because a single regression model trained on this target would be dominated by the large number of zero and near-zero snowfall days.

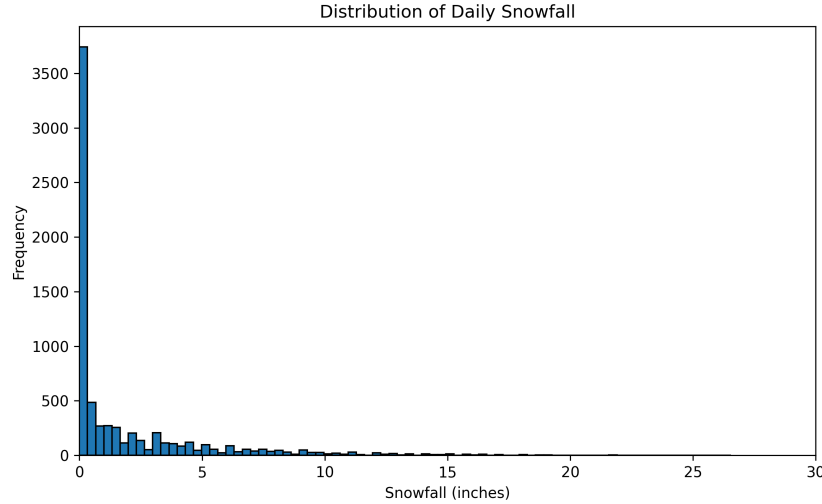


Figure 3: Distribution of daily snowfall observations at Grand Targhee.

The confusion matrix provided detailed information on classifier performance across the 1,426 test day set. The model correctly identified 1,012 no-snow days and 207 snow days, demonstrating strong overall classification performance across both classes. The 73 false positives represent days where snowfall was predicted but did not occur, while the 134 false negatives represent actual snowfall events missed by the classifier. The relatively low false positive count reflects the improved precision achieved using the 0.72 classification threshold, while the false negatives highlight the tradeoff between precision and recall. Lowering the classification threshold would likely increase recall but at the cost of more false snowfall predictions. For ski resort planning, the 134 missed snow days represent the most significant forecasting errors as they could reduce preparation time.

The actual vs. predicted scatter plot revealed several important forecasting behaviors of the two-stage XGBoost framework. A clear positive relationship existed between observed and predicted snowfall amounts, showing that the model was effective at capturing overall snowfall trends and moderate snowfall events. However, the scatter plot also revealed sources of forecasting error. The horizontal group of points along the bottom axis represented snowfall events missed by the classifier in which snowfall occurred but the model predicted zero snowfall (false negatives). The plot also showed some overprediction for low-to-moderate snowfall amounts. Some extreme snowfall events were considerably underestimated by the model; the largest event (26 inches) was predicted at 9 inches. This behavior is consistent with the higher RMSE observed for the XGBoost framework since rare, large snowfall events contribute disproportionately to squared prediction error and remain difficult to forecast accurately.

4 Discussion

4.1 Benefits of a Two-Stage Model

Because daily snowfall accumulation is zero on a large proportion of days throughout the year, single-stage models are heavily impacted by the heavy rightward skew of the snowfall per day histogram (see histogram above). This took away from the model's ability to be able to predict the level of snowfall. By separating this reasoning into two stages, we allow the model to fully learn patterns when they matter most, yielding the most accurate predictions. After considering what outputs would actually be useful, we determined the model should work by first using a classifier

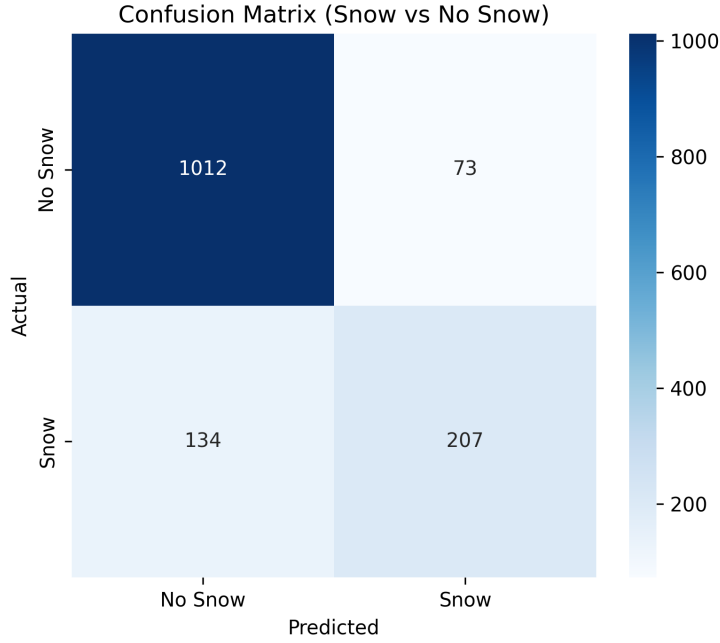


Figure 4: Confusion matrix for snowfall occurrence classification using the two-stage XGBoost framework. The classifier achieved strong overall performance with relatively few false positive snowfall predictions.

to predict whether or not snowfall would occur at all, and if so, predicting the magnitude of said snowfall with a conditional regressor. This separation of forecasting tasks allows the regression stage to focus specifically on learning the relationships associated with storm conditions, improving the utility of its predictions. Since the magnitude of snowfall depends on many nonlinear relationships between factors like weather patterns, temperature, and precipitation, deriving a useful prediction while also considering the many days with no snowfall would have been a tremendous waste of resources.

Although the Random Forest baseline achieved a slightly lower RMSE than the two-stage XGBoost model, the XGBoost system provided several additional forecasting capabilities beyond raw regression accuracy. By separating snowfall occurrence prediction from snowfall amount regression, the forecasting framework was able to produce probabilistic snowfall occurrence estimates in addition to snowfall magnitude predictions.

4.2 Importance of Temporal Features

The inclusion of Temporal Features in our model was critical to improving forecasting accuracy. Fluctuating weather conditions are a major contributing factor for predicting snowfall magnitude, and data from previous days/months/years is able to be used to predict these conditions. In order to guarantee these temporal relationships, we enhanced our data for snowfall, snow depth, temperature and precipitation with lag variables and rolling-window statistics. These additions allow the model to recognize storm patterns and other natural phenomena that are critical to predicting snowfall magnitude. Specifically, long-term trends such as temperature and precipitation histories can greatly help the model tell the difference between short term storm conditions and seasonal trends, enabling it to better predict exceptional snowfall, which is arguably one of the most important features of the model. Without these features, our model would be substantially more rigid with its learning, vastly decreasing its usefulness and reliability. These features helped improve the model's ability to capture complex snowfall behavior relative to simpler baseline forecasting approaches.

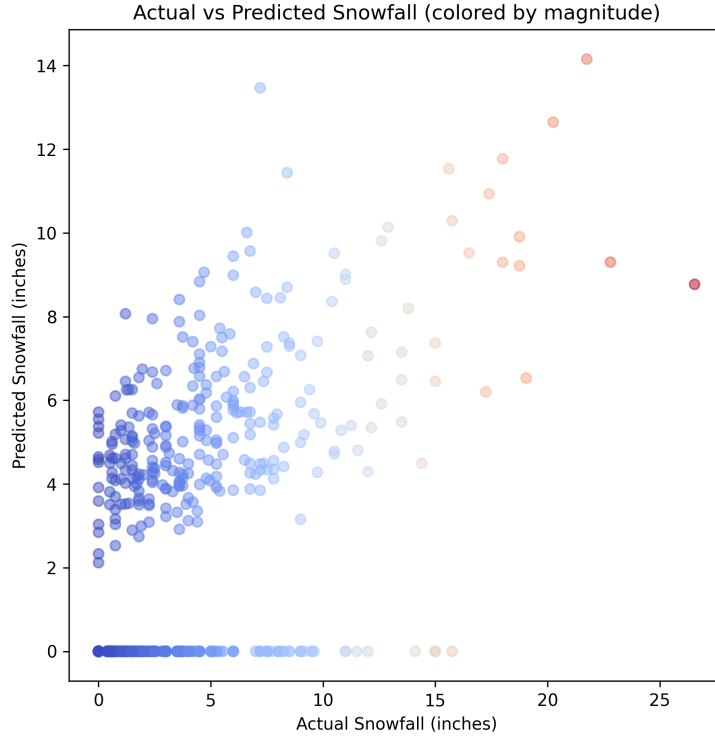


Figure 5: Actual versus predicted snowfall amounts for the two-stage XGBoost model.

4.3 XGBoost

XGBoost proved to be an effective model for capturing nonlinear snowfall behavior and supporting probabilistic two-stage forecasting. This was largely because the features that contribute to overall snowfall magnitude have many nonlinear relationships with one another, since they are all environmental factors. Additionally, when considering the typical conditions at ski resorts, the around-freezing temperatures can frequently create a dramatic difference in snowfall with only a few degree change in temperature, since that snow could turn into rain. The nature of tree-based ensemble methods like XGBoost to iteratively partition the feature space using regions abstracted from similar predictive behavior makes them particularly well suited to model sensitive circumstances like those at ski resorts. XGBoost’s ability to adapt to changing environmental trends through temporal features also enabled it to capture many more complicated dependencies that would be outright overlooked by a linear model. The model also utilized our boosting framework to progressively correct residual forecasting errors from previous trees, substantially refining predictive accuracy for unexpected and unique weather events. Overall, these traits allowed XGBoost to engage much more deeply with our provided data than a baseline model, since the nature of our data is noisy, seasonal-trend oriented, and contains many nonlinear relationships between features.

4.4 Real-World Advantages

Over the course of development, our forecasting framework has displayed many real, practical implications for ski resort logistics, in addition to predictive accuracy. Having access to our deep, reliable forecasting could greatly assist the management of a ski resort with many important decisions, such as staff allocation, trail preparation, and availability timelines. Specifically, our model’s ability to predict exceptional weather conditions can provide extremely valuable insights that ensure a resort is prepared for periods of increased demand, and able to guarantee the safety of all patrons. The

forecasting framework demonstrated the ability to identify important snowfall conditions that may be useful for ski resort operational planning. Overall, this model demonstrates how machine learning methods can make better use of data, and utilize it to make insightful predictions beyond those provided by traditional deterministic forecasting methods. With the effectiveness and advantages of this model, it is clear that this same concept could be applied to all manner of different applications that involve weather-related forecasting.

5 Limitations

5.1 Regional Scope

One limitation of this project was the relatively limited scope of data used for training and testing. Specifically, our models were trained using historical data from a single SNOTEL region, Grand Targhee, due to the sheer scale of data. While our model was able to effectively learn from this data set, it was never exposed to the massively varying environmental factors found in other regions. Since skiing occurs in many different regions, the inclusion of many more SNOTEL datasets could greatly increase the applicability of our model for many ski resorts not in the Grand Targhee region going forward.

5.2 Short-Term Environmental Variability

Another limitation of this project was the level of temporal detail in our data relating to environmental factors. Our forecasting system utilized daily aggregated weather reports rather than higher-frequency measurements, such as hourly. As with regional scope, we approached training in this manner due to the sheer scale of data that would need to be processed. Due to the nature of the environment, changes in factors like temperature and precipitation can occur quickly, cascading into rapidly evolving conditions that would be useful to know about in advance. Adaptation of the model to hourly data could further enhance the practical benefit of the system for ski resorts, since it could potentially predict unsafe or wavering conditions. These benefits make this improvement worth pursuing in the future.

5.3 Advanced Features

Another limitation of this project stems from the meteorological methodology behind the data that we utilized to train our model. This system heavily relies on SNOTEL weather station observations, and their connection to our engineered temporal features. While this data was certainly enough to obtain a reliable predictor, a vast number of more advanced meteorological data exists, and much of it could be applied to our system. Specifically, data sources such as satellite imagery, radar measurements, and pressure maps could contain trends that our model was unable to train on with the current data layout. As a result, our model may miss important storm development patterns or global atmospheric conditions that could have a substantial impact on predictions. These features were not included in our project due to the complexity and scale of their data, but investing time into making our current system compatible with these advanced technologies could have huge benefits for prediction accuracy, making it a worthwhile investment going forward.

6 Conclusion

Throughout this project, we were able to successfully train a two-stage XGBoost model to make reliable snowfall predictions using a wide variety of environmental features from SNOTEL historical data. By separating snowfall occurrence prediction from snowfall amount regression, the forecasting framework was able to effectively model snow and no-snow conditions along with snowfall intensity patterns. Engineered temporal and seasonal features including lag variables, rolling-window statistics, interaction features, and cyclical seasonal encodings further improved the model's ability to capture recurring weather behavior and storm persistence patterns.

The results demonstrated that machine learning approaches can effectively model complex relationships between precipitation, temperature, snow depth, and seasonal atmospheric conditions. While extreme snowfall events remained difficult to accurately predict, the two-stage forecasting framework

achieved strong performance and competitive results relative to baseline regression models. Additionally, this project demonstrated the importance of temporal weather relationships and probabilistic snowfall prediction for environmental forecasting applications.

Overall, this work demonstrates how machine learning forecasting systems can make use of engineered temporal and environmental weather data to improve snowfall prediction performance. The forecasting framework developed in this project could provide useful insights for ski resort planning, staffing, and weather-related decision making. Future work could further improve on this work by expanding the model across multiple ski resort regions and incorporating additional meteorological data sources.

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